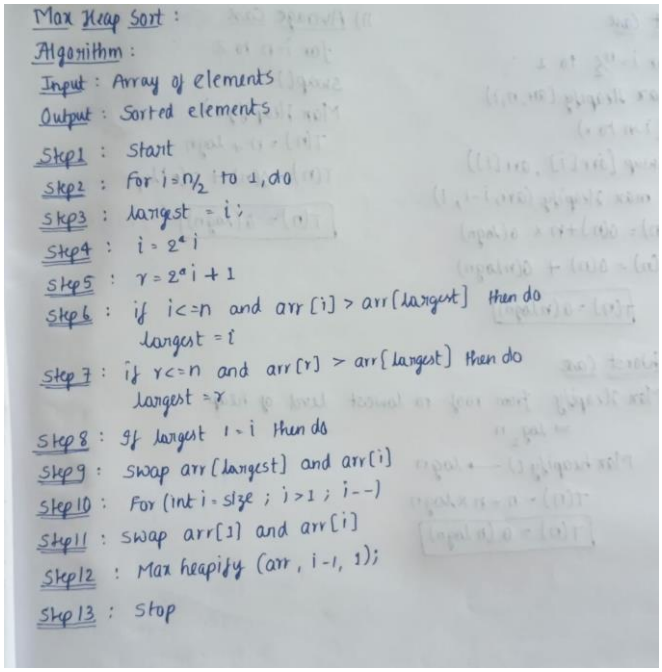
PRACTICAL-03

AIM: Implementation of max-heap sort algorithm.

1.) MAX HEAP

ALGORITHM:



PROGRAM:

```
#include <iostream>
using namespace std;

void maxheapify(int arr[], int size, int i)
{
   int largest = i;
   int l = 2 * i;
   int r = 2 * i + 1;

   if (l <= size && arr[l] > arr[largest])
   {
```

```
        largest = l;
    }

    if (r <= size && arr[r] > arr[largest])
    {
        largest = r;
    }

    if (largest != i)
    {
        swap(arr[largest], arr[i]);
        maxheapify(arr, size, largest);
    }
}
void heapsort(int arr[], int size)
{
    for (int i = size / 2; i >= 1; i--)
    {
        maxheapify(arr, size, i);
    }
    for (int i = size; i > 1; i--)
    {
        swap(arr[1], arr[i]);
        maxheapify(arr, i - 1, 1);
    }
}
int main()
{
    int n;
    cout << "Enter number of elements: ";
    cin >> n;
    int arr[n + 1];
    cout << "Enter elements: ";
    for (int i = 1; i <= n; i++)
    {
        cin >> arr[i];
```

```
    }
    heapsort(arr, n);
    cout << "Sorted array: ";
    for (int i = 1; i <= n; i++)
    {
        cout << arr[i] << " ";
    }
    return 0;
}
```

OUTPUT:

```
Enter number of elements: 5
Enter elements: 64 34 25 12 22
Sorted array: 12 22 25 34 64
```

TIME COMPLEXITY:

